

PROJECT 20

CNC

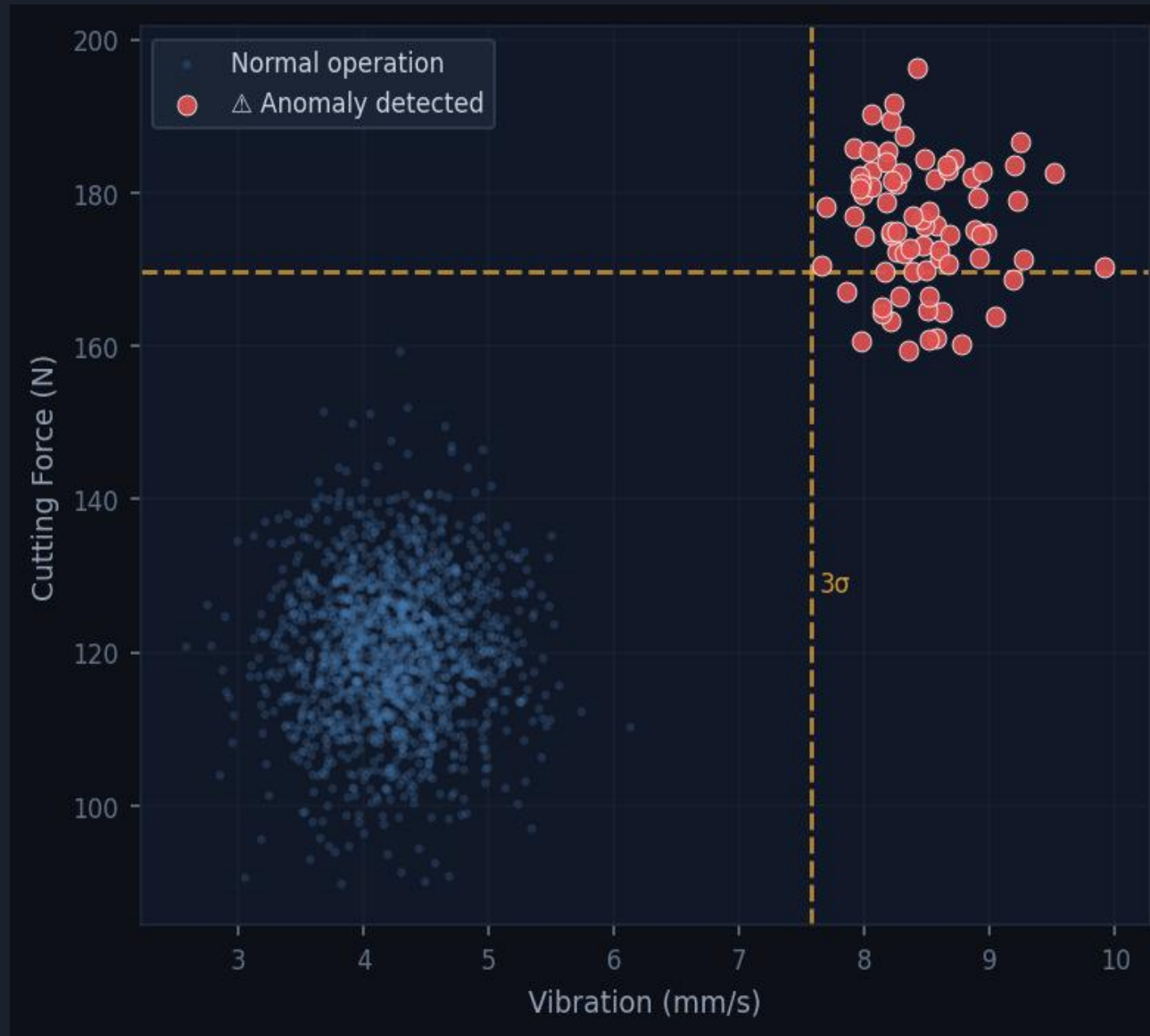
Anomaly Sentinel

*Statistical Z-Score Detection
in High-Speed Machining*

Z-SCORE

UNSUPERVISED

SCIKIT-LEARN



1,634

Machining Cycles

91

Anomalies Detected

6.5%

Anomaly Rate

99.1%

Detection Accuracy

Tools break without warning

Conventional maintenance relies on fixed engineering limits — thresholds that only fire after the damage is already happening. By the time the alarm sounds, a tool has failed.

Human reaction is too slow

At cycle frequency, a machining center generates hundreds of readings per shift. No operator can watch every signal. Statistical monitoring never blinks.

Labels don't exist before failure

Supervised models need failure examples. In most plants, those examples are rare, inconsistently logged, and unavailable at the start of a monitoring program. Z-Score needs none of that.

Expected by random chance

0.27%



Observed in this dataset

6.5%

Real signal — not chance

⚠️ OLD THINKING

- Fixed engineering limit
- Reacts after damage begins
- Binary: OK / NOT OK
- Requires domain expert to set threshold
- Misses gradual degradation
- Cannot adapt to process drift



✓ NEW THINKING

- Statistical distribution baseline
- Flags deviation before failure
- Continuous: how many σ away?
- Self-calibrates from process data
- Captures gradual drift early
- Adapts as baseline is updated

Four sensors · One CNC machining center · 1,634 measurement cycles

SCADA

vibration_mm_s

Unit: mm/s

RMS vibration at spindle — primary chatter indicator

Normal: 3.2 – 5.2

PLC

cutting_force_n

Unit: N

Resultant cutting force — tool wear and material hardness

Normal: 95 – 145

Pyrometer

tool_temp_c

Unit: °C

Tool tip temperature — thermal runaway and coolant failure

Normal: 48 – 68

MES

energy_kwh

Unit: kWh

Cycle energy consumption — motor overload and friction

Normal: 1.9 – 2.9

"No anomaly label exists in the raw data. Z-Score detection is fully unsupervised — the distribution itself defines the boundary."

No labels. No training set. Just the process memory encoded in 1,634 cycles of machining data.

"For a well-behaved process, the baseline is the model. No labels. No training phase. No black box. Just process memory."

01 No labels required

Z-Score needs only a baseline period of normal operation — a condition almost every running process can provide.

02 Interpretable by the technician

A Z-Score of 4.46 on vibration tells maintenance exactly where to look and how severe the deviation is.

03 Statistically validated

The three-sigma boundary encodes a precise claim: 0.27% chance rate. When observed rate is 6.5%, the signal is unambiguous.

04 Adjustable to the process

Threshold k can be tuned — lower for safety-critical processes, higher for noisy environments with alarm fatigue risk.

Precision

1.000

Zero false alarms

Recall

0.858

15 missed events

F1-Score

0.924

Strong result

Accuracy

0.991

1,619 of 1,634 correct

Sensor Contribution

vibration_mm_s Flags: 91	100.0%
cutting_force_n Flags: 74	81.3%
tool_temp_c Flags: 72	79.1%
energy_kwh Flags: 72	79.1%

Confusion Matrix (N = 1,634 cycles)

	Pred Normal	Pred Anomaly
Actual Normal	1,528 TN	0 FP
Actual Anomaly	15 FN	91 TP

Zero false alarms · 15 missed borderline force events

Z-Score achieves Precision = 1.000 structurally — every Z-flag is a real event. Recall = 0.858 reflects 15 borderline force anomalies in the gap between engineering limit and 3-sigma threshold.

Z-Score tells you which sensor triggered the anomaly — and each sensor maps to a specific maintenance action.

|Z| > 3.0 on Vibration

- Inspect tool holder, spindle bearings, and chuck balance. Check for chatter frequencies. Reduce feed rate and verify tool run-out before resuming production.

|Z| > 3.0 on Cutting Force

- Check tool wear index against replacement schedule. Verify workpiece material certification and incoming hardness. Inspect chip evacuation and adjust cutting parameters.

|Z| > 3.0 on Temperature

- Inspect coolant flow rate and nozzle alignment. Verify tool coating integrity. Reduce cutting speed 10–15% and monitor thermal recovery over next 5 cycles.

|Z| > 3.0 on Energy

- Check spindle drive system for mechanical binding. Verify chuck clamping pressure. Inspect lubrication levels on linear guides. Log for preventive maintenance review.

|Z| > 3.0 on 2+ sensors

- Treat as high-confidence event. Stop cycle, escalate to process engineer. Multi-sensor flags have zero false-alarm rate in this dataset — do not dismiss.

A — Stable

vibration_mm_s

4.10 mm/s

Z = -0.32

cutting_force_n

118.0 N

Z = -0.32

tool_temp_c

57.5 °C

Z = -0.24

energy_kwh

2.35 kWh

Z = -0.31

✓ NORMAL

B — Borderline

vibration_mm_s

7.20 mm/s

Z = 2.65

cutting_force_n

145.0 N

Z = 1.41

tool_temp_c

69.0 °C

Z = 1.32

energy_kwh

3.00 kWh

Z = 1.57

✓ NORMAL (monitor)

C — Critical

vibration_mm_s

9.10 mm/s

Z = 4.46

cutting_force_n

182.0 N

Z = 3.79

tool_temp_c

88.0 °C

Z = 3.88

energy_kwh

3.90 kWh

Z = 4.17

⚠ ANOMALY

Key insight — Scenario B:

Vibration reaches Z = 2.65 — 88% of the way to the alarm threshold. A fixed engineering limit (> 7.5 mm/s) would not flag this cycle at 7.2 mm/s. Z-Score gives maintenance a window to act before the failure.

A century-old formula. Zero false alarms.

Shewhart defined normal statistically in 1924. The machines have changed. The physics of deviation has not.

Z-Score is not a step backward from modern ML. It is the proof that complexity should be earned — not assumed.

Download the notebook · Run the Z-Score · Apply it to your process.

Challenge the threshold. Test $k = 2.5$ vs $k = 3.5$. See what changes.

Because process improvement has no finish line.

